
Online Examination Report

Date:	02.08.2014
Attempts:	1
Examinee:	Okeng, Japheth
Date of birth:	
Subject:	020-AGK
Result:	0 %
Time used:	00:00:08 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2766	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
2	2739	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	2719	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	2437	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
5	597	21.2.3.1	Design and construction	0,00	1,00
6	1350	21.2.4	Fuselage, landing gear, doors, floor, etc	0,00	1,00
7	2112	21.3.2	Hydraulic systems	0,00	1,00
8	2179	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
9	1335	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
10	2049	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
11	468	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
12	2235	21.9.1	General, definitions, basic applications: etc	0,00	1,00
13	1998	21.9.2	Batteries	0,00	1,00
14	2277	21.11.1	Basic principles	0,00	1,00
15	2363	21.11.1	Basic principles	0,00	1,00
16	2329	21.12.2	Fire protection systems	0,00	1,00
17	1531	21.12.2.2	Fire detection	0,00	1,00
18	584	21.13	OXYGEN SYSTEMS	0,00	1,00
19	1623	21.13	OXYGEN SYSTEMS	0,00	1,00
20	2334	21.13	OXYGEN SYSTEMS	0,00	1,00
21	4026	22.	INSTRUMENTATION	0,00	1,00
22	2822	22.	INSTRUMENTATION	0,00	1,00
23	4129	22.	INSTRUMENTATION	0,00	1,00
24	3090	22.	INSTRUMENTATION	0,00	1,00
25	1779	22.1.1	Pressure gauge	0,00	1,00
26	313	22.2.1	Pressure measurement	0,00	1,00
27	1884	22.2.1.1	Definitions	0,00	1,00
28	1946	22.2.4	Altimeter	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	1053	22.2.4	Altimeter	0,00	1,00
30	1690	22.2.6	Airspeed indicator (ASI)	0,00	1,00
31	1320	22.2.7	Mach meter	0,00	1,00
32	1294	22.2.7	Mach meter	0,00	1,00
33	1198	22.2.7	Mach meter	0,00	1,00
34	1389	22.2.7	Mach meter	0,00	1,00
35	1646	22.3	MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	0,00	1,00
36	1522	22.4.1	Gyroscope: basic principles	0,00	1,00
37	629	22.6.3	Flight Director : design and operation.	0,00	1,00
38	1422	22.6.3	Flight Director : design and operation.	0,00	1,00
39	327	22.12.3	Stall Warning Systems (SWS)	0,00	1,00
40	1802	22.12.8	Radio Altimeter	0,00	1,00
Total:0%				0,00	40,00

Attachment 2: List of result by topic

Licence

Date: 02.08.2014

Examinee: Okeng, Japheth

Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	0,00	20,00	0,00
21.2. AIRFRAME	3	0,00	3,00	0,00
21.3. HYDRAULICS	1	0,00	1,00	0,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	3	0,00	3,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	1	0,00	1,00	0,00
21.9. ELECTRICS	2	0,00	2,00	0,00
21.11. TURBINE ENGINES	2	0,00	2,00	0,00
21.12. PROTECTION AND DETECTION SYSTEMS	2	0,00	2,00	0,00
21.13. OXYGEN SYSTEMS	3	0,00	3,00	0,00
22. INSTRUMENTATION	20	0,00	20,00	0,00
22.1. SENSORS AND INSTRUMENTS	1	0,00	1,00	0,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	9	0,00	9,00	0,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	1	0,00	1,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	1	0,00	1,00	0,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	2	0,00	2,00	0,00
22.12. ALERTING SYSTEMS, PROXIMITY SYSTEMS	2	0,00	2,00	0,00
Total	40	0,00	40,00	0,00

Candidate: Japheth Okeng
Date of birth: --

Subject: 020-AGK
Date: 2014-08-02

1

The maximum cumulative time that an emergency locator transmitter may be operated before the rechargeable battery must be recharged is

☐ 30 minutes.

☒ 60 minutes.

☐ Regulations5 minutes.

2

Detonation in an aircraft engine is most likely to occur whenever the

- ☒ **engine is operated under conditions which cause the fuel mixture to burn instantaneously.**
- ☐ **fuel being used is of a higher grade than recommended by the engine manufacturer.**
- ☐ **fuel/air ratio is such that the mixture burns extremely slow.**

3

Which statement is true regarding fouling of the spark plugs of an aircraft engine?

- ☐ Excessive heat in the combustion chamber of a cylinder causes oil to form on the center electrode of a spark plug and this fouls the plug.
- ☐ Carbon fouling of the spark plugs is caused primarily by operating an engine at excessively high cylinder head temperatures.
- ☒ Spark plug fouling results from operating with an excessively rich mixture.

4

What is the reason for variations in geometric pitch along a propeller or rotor blade?

- ☐ It permits a relatively constant angle of incidence along its length when in cruising flight.
- ☐ It prevents the portion of the blade near the hub or root from stalling during cruising flight.
- ☒ It permits a relatively constant angle of attack along its length when in cruising flight.

5

Among the different types of aircraft structures, the shell structures efficiently transmit the: 1. normal bending stresses 2. tangent bending stresses 3. torsional moment 4. shear stresses The combination regrouping all the correct statements is :

☐ 2, 3, 4

☐ 1, 3, 4

☒ 1, 2, 3

☐ 1, 2, 4

6

When the door operation of a transport airplane equipped with evacuation slides is controlled from the outside, the slide:

- ☒ **is disarmed automatically.**
- ☐ **unfolds but does not become inflated.**
- ☐ **becomes inflated in its packboard thus preventing its unfolding.**
- ☐ **unfolds and becomes inflated.**

7

Hydraulic fluids used in systems of large modern airliners are:

☒ **Phosphate ester base fluids.**

☐ **Vegetable base fluids.**

☐ **Mineral base fluids.**

☐ **Water base fluids.**

8

If the profile grooves or the tread of a new aircraft tyre are worn, the tyre can be:

- ☒ Repaired several times.
- ☐ Never repaired.
- ☐ Used on the nose wheel only.
- ☐ repaired once.

9

Thermal plugs are installed in:

- ☐ cargo compartments.
- ☐ fire warning systems.
- ☒ wheel rims.
- ☐ cabin windows.

10

A torsion link assembly is installed on the landing gear to :

- ☐ control the wheels.
- ☒ avoid rotation of the piston rod relative to the gear oleo strut.
- ☐ absorb the spring tension.
- ☐ lock the landing gear.

11

If the pressure in the cabin tends to become lower than the outside ambient air pressure the :

- ☐ outflow valve open completely.
- ☒ negative pressure relief valve will open.
- ☐ negative pressure relief valve will close
- ☐ air cycle machine will stop.

12

In an aeroplane utilising a constant frequency AC power supply, DC power is obtained from a :

- ☐ rotary converter.
- ☐ static inverter.
- ☐ 3 phase current transformer unit.
- ☒ Transformer Rectifier Unit.

13

A test to assess the state of charge of a lead-acid battery would involve :

- ☒ comparing the "on-load" and "off-load" battery voltages.
- ☐ checking the discharge current of the battery "on-load".
- ☐ checking the battery voltage "off-load".
- ☐ checking the level of the electrolyte.

14

In jet aeroplanes the 'thermal anti-ice system' is primary supplied by

- ☐ turbo compressors.
- ☐ ram air, heated via a heat exchanger.
- ☒ bleed air from the engines.
- ☐ the APU.

15

The Engine Pressure Ratio (EPR) is the ratio of:

- ☐ the total turbine outlet pressure to the total compressor outlet pressure.
- ☒ the total turbine outlet pressure to the total compressor inlet pressure.
- ☐ the total turbine inlet pressure to the total compressor outlet pressure.
- ☐ the total turbine inlet pressure to the total compressor inlet pressure.

16

An aircraft whose maximum approved configuration for passenger seats is 10 seats must be equipped with:

- ☐ two fire-extinguishers in the cockpit and two fire-extinguishers in the passenger cabin.
- ☐ one fire-extinguisher in the cockpit and two fire-extinguishers in the passenger cabin.
- ☒ one fire-extinguisher in the cockpit and three fire-extinguishers in the passenger cabin.
- ☐ three fire-extinguishers in the passenger cabin only.

17

Continuous loop fire detector systems operate on the principle that an increase in temperature produces:

- ☐ *an increase in resistance*
- ☐ *a decrease in pressure*
- ☒ *a decrease in resistance*
- ☐ *a decrease in the reference current*

18

A diluter demand oxygen regulator :

- ☒ **delivers oxygen flow when inhaling.**
- ☐ **is only recommended for use with smoke in the cockpit.**
- ☐ **mixes air and oxygen in a passenger oxygen mask.**
- ☐ **delivers oxygen flow only above FL 100.**

19

When air is compressed for pressurization purposes, the percentage oxygen content is:

- ☐ decreased.
- ☐ dependent on the degree of pressurisation.
- ☒ unaffected.
- ☐ increased.

20

In a pressurized aircraft, the first-aid (therapeutic) oxygen is designed to:

- ☐ protect all the occupants against the effects of accidental depressurisation.
- ☐ protect the flight crew and cabin attendants against fumes and noxious gases.
- ☒ give medical assistance to passengers with pathological respiratory disorders.
- ☐ protect certain passengers, and is only carried on board for these people.

21

Which is the maximum tolerance for the VOR indication when the CDI is centered and the aircraft is directly over the airborne checkpoint?

- ☒ Plus or minus 6 degrees of the designated radial.
- ☐ Plus or minus 4 degrees of the designated radial.
- ☐ Plus 6degrees or minus 4 degrees of the designated radial.

22

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☐ **airspeed indicator may indicate less than normal.**
- ☒ **altimeter may indicate an altitude higher than the actual altitude being flown.**
- ☐ **altimeter may indicate an altitude lower than the actual altitude being flown.**

Candidate: Japheth Okeng
Date of birth: --

Subject: 020-AGK
Date: 2014-08-02

23

(Refer to Figure 85.) What is the minimum rate climb per NM to 9,000 feet required for the WASH2 WAGGE Departure?

☐ 750 feet.

☒ 400 feet.

☐ 875 feet.

See Attachments:

1. instrument_85.jpg

figure 85

(Attachment No.1)

24

Identify taxi leadoff lights associated with the centerline lighting system.

- ☐ Alternate green and yellow lights curving from the centerline of the runway to the edge of the taxiway.
- ☐ Alternate green and yellow lights curving from the centerline of the runway to the centerline of the taxiway.
- ☒ Alternate green and yellow lights curving from the centerline of the runway to a point on the exit.

25

If a manifold pressure gauge consistently registers atmospheric pressure, the cause is probably:

- ☐ a very high atmospheric pressure.
- ☐ an aneroid capsule which is not properly calibrated.
- ☒ leak in pressure gauge line.
- ☐ a very low atmospheric pressure.

26

If while in level flight, it becomes necessary to use an alternate source of static pressure vented inside the aircraft, with the cabin pressure being lower than static, the following variations in instrument indication would be expected:

- ☒ the altimeter will read higher than normal, airspeed greater than normal and the vertical-speed indicator will momentarily show a climb;
- ☐ the altimeter will read lower than normal, airspeed greater than normal and the vertical-speed indicator will momentarily show a climb and then a descent.
- ☐ the altimeter will read higher than normal, airspeed will not change and the vertical-speed indicator will momentarily show a descent;

27

Concerning the pitot and static system, the static pressure error:

- ☐ *is a direct effect of a blockage of the static port.*
- ☒ *is caused by disturbed airflow around the static ports.*
- ☐ *affects the alternate static port only.*
- ☐ *is a direct effect of heating of the static ports.*

Candidate: Japheth Okeng
Date of birth: --

Subject: 020-AGK
Date: 2014-08-02

28

During a pre-flight check the following details were noted: Airfield Elevation (4800 ft)- Apron Elevation (4780 ft)- Height of static vent above ground (25 ft)- Altimeter reading (QFE) (35 ft)

☒ 30 ft underread.

☐ 5 ft underread

☐ 30 ft underread

29

In a servo-assisted altimeter, the secondary winding from the E bar provides an electrical current direct to the:

- ☒ amplifier
- ☐ AC input lines of the primary winding
- ☐ two phase motor
- ☐ capsule stack

30

VFE is the maximum speed:

- ☒ *with the flaps extended.*
- ☐ *with the flaps extended in the take-off configuration.*
- ☐ *at which the flaps can be operated when flying in turbulences.*
- ☐ *with the flaps extended in the landing configuration.*

31

The percentage error due to the positioning of the static vent is:

- ☐ greater in the Machmeter than in the ASI
- ☐ either greater or smaller when comparing the ASI and the Machmeter,
- ☐ greater in the ASI than in the Machmeter
- ☒ the same in the Machmeter as in the ASI

Candidate: Japheth Okeng
Date of birth: --

Subject: 020-AGK
Date: 2014-08-02

32

If an aircraft climbs at a constant IAS in the standard atmosphere, the Mach number will:

- ☒ increase
- ☐ remain constant
- ☐ remain constant unless the aircraft flies through an inversion
- ☐ decrease

33

A machmeter does not suffer from:

- ☒ compressibility error
- ☐ instrument error
- ☐ error due to blockage of the static vent
- ☐ position error

34

In a combined Mach/airspeed indicator the purpose of the barber's pole is to indicate:

- ☐ Vmo, and this is a fixed value at all altitudes
- ☐ Vmo, and this value decreases at high altitudes
- ☐ the maximum safe airspeed with the flaps/slats extended
- ☒ Vmo, and this value increases at high altitudes

35

The rotor of the DGI spins up and away from the pilot when 090 is indicated. The latitude compensation nut situated on the near right hand side of the inner gimbal from the gyro axis, has been set to give zero drift on the ground at the equator. To compensate for earth rotation at 30 S the latitude compensating nut:

☐ must be adjusted inwards;

☐ is not adjusted since the latitude nut can only be used to correct for apparent wander in the northern hemisphere.

☒ must be adjusted outwards;

36

The angle of tilt of the rate gyro in a turn indicator is due to:

- ☐ the force in the horizontal plane generated by secondary precession balancing the tilt caused by primary precession;
- ☒ primary precession which is generated by an aeroplanes rate of turn.
- ☐ the tension of the control spring opposing the angular tilt of the gyro;

37

Considering a flight director of the command bars type:

- ☐ *the horizontal bar is associated with the roll channel.*
- ☐ *the vertical bar may be associated with the pitch channel.*
- ☒ *the horizontal bar is associated with the pitch channel.*
- ☐ *the vertical bar is associated with the pitch channel.*

38

The horizontal command bar of a flight director:

- 1 - repeats the position information given by the ILS in the horizontal plane
- 2 - repeats the position information given by the ILS in the vertical plane
- 3 - gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination that regroups all of the correct statements is:

☐ 2, 3.

☐ 2.

☒ 3.

☐ 1, 3.

39

The stall warning computer uses the following data:

- 1- mass of the aeroplane
- 2- angle of attack
- 3- wing flap deflection
- 4- position of the landing gear
- 5- TAT
- 6- pressure altitude

The combination that regroups all of the correct statements is:

☐ 1, 2, 3, 4, 5, 6

☒ 2, 3

☐ 1, 2, 3, 4

☐ 2, 3, 5

40

The radio altimeter uses the following wavelengths:

☐ *myriametric.*

☐ *metric.*

☐ *millimetric*

☒ *centimetric.*

Attachment No. 1

Questions 23

figure 85

instrument_85.jpg

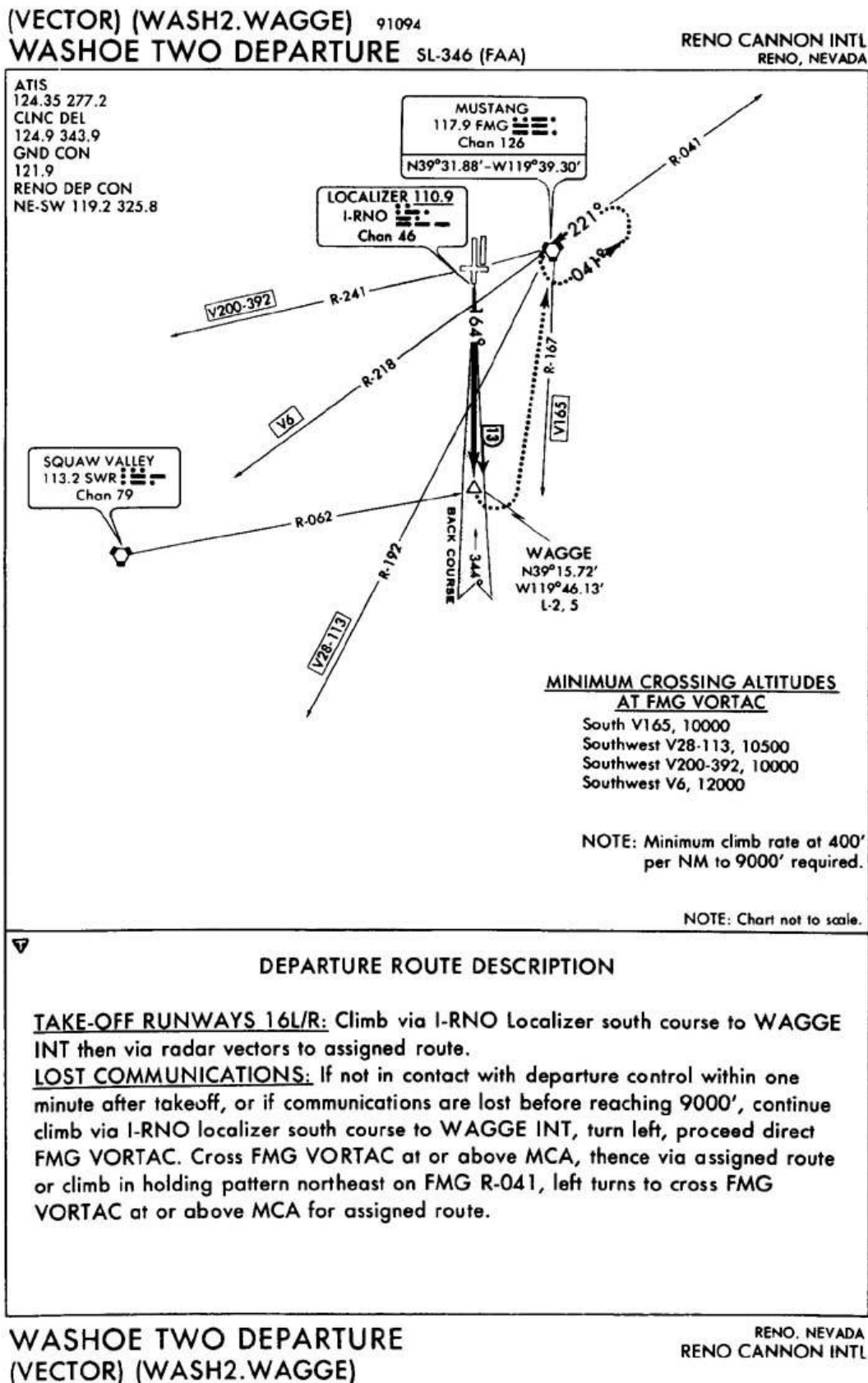


Figure 85. WASHOE TWO DEPARTURE © ASA